THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Chapter 10 — Phase III: The Grammar of Structure

Self-Study Curriculum

What a Dynamical System Is

- Anything that runs forward in time by a rule: a population, a drug clearing, a gene switching on.
- The field your mechanistic instinct already lived in, unnamed.
- The flagship example (decay = populations = capacitors) is a fact about *change*, not algebra.

Topology gave a static stage. Dynamics puts a flow on it.

State and State Space

- **State**: the minimal numbers that fix the entire future from *now*.
- Isotope: one number (amount). Pendulum: two (angle *and* angular velocity).
- Minimal = the present is a sufficient summary of the past (Markov property).
- The set of all states is the **state space** (phase space).

The Flow

- The rule $dx/dt = f(x)$ is a *vector field*: an arrow at every point.
- A trajectory = drop a marble, follow the arrows.
- Stop studying one history; study the whole field of all histories at once.

State space = the canvas of Chapter 8, now set in motion.

Fixed Points: The Skeleton

- A **fixed point** is where $f(x) = 0$: velocity vanishes, the system stays.
- Equilibria are the skeleton of the flow; find them first.
- Stability is NOT read off the equilibrium *value*.
- It is a property of the flow *around* the point: do neighboring arrows converge or flee?

Three Kinds of Equilibrium

- **Stable**: pulls nearby states in. Marble in a bowl.
- **Unstable**: pushes nearby states away. Marble on a dome.
- **Saddle**: stable some directions, unstable others. Marble on a mountain pass.
- Same location can be stable or unstable; the flow around it decides.

Logistic Worked Example

- $dN/dt = rN(1 - N/K)$. Fixed points: $N = 0$ and $N = K$.
- Small N: arrows point right (grow). Above K: arrows point left (fall back).
- Extinction (0) is unstable; carrying capacity (K) is stable.
- Read straight off the arrows, without solving the equation.

Linearization

- Near x*, write state as x* $+ \delta$, with δ small.
- Taylor: f(x* $+ \delta) =$ f(x*) $+$ f'(x*)$\cdot\delta +$ (order δ^2 and smaller).
- At a fixed point f(x*) $= 0$, so dδ/dt $\approx$ f'(x*)$\cdot\delta$.
- Local evolution is *linear*; the coefficient is the **Jacobian** matrix.

Eigenvalues Are Rate Constants

Jacobian eigenvalue	What the flow does
negative real part	decays, stable
positive real part	grows, unstable
nonzero imaginary part	rotates, oscillation

- The eigenvalue is the *rate constant* of the flow (its analytic role, Chapter 3).

Why Linear Methods Work

- Not because the world is linear. Because the world is *smooth*.
- Taylor: a smooth system, viewed closely near equilibrium, *is* linear.
- That is the deepest reason linear algebra is disproportionately powerful.

Linearity is the first-order shadow every smooth system casts near a fixed point.

Linearity Lives in the Canvas

- Two senses of "linear": $dN/dt = rN$ already superposes in N; plotting log N straightens the *graph*, a different thing.
- Your own intuition, made exact: linearity is system-plus-canvas, not the system alone.
- The displacement δ *is* the right canvas near equilibrium: there the *rule* superposes, whatever f was.
- Error, stability margin, and perturbation size are one quantity: the size of δ.

Multistability

- Two or more stable fixed points; outcome depends on where you start.
- State space splits into **basins of attraction** separated by ridges.
- Cell-fate decisions, genetic toggle switches, hysteresis: memory in the landscape.
- A linear system has one equilibrium, so it can offer only one destiny.

Oscillation

- A **limit cycle**: a closed loop trajectories spiral onto. A stable *rhythm*.
- Circadian clocks, heartbeats, predator-prey, gene oscillators.
- Linear systems oscillate only at a knife's edge, amplitude set by the kick.
- Born in a **Hopf bifurcation**: a complex eigenvalue pair crosses into positive real part.

Bifurcation: The Meaning of Threshold

- A *qualitative* reorganization of the flow as a parameter crosses a critical value.
- A fixed point appears, vanishes, or swaps stability; behavior changes all at once.
- This is the “threshold” a linear (proportional-everywhere) model structurally cannot carry.
- It is Chapter 14’s “structural misspecification” seen from the inside.

Chaos

- Deterministic yet unpredictable: an exact rule, no randomness.
- Nearby trajectories separate exponentially (sensitive dependence).
- Short rule, incompressible long-range behavior.

"We know the equations" and "we can predict the future" are different claims (Chapter 15).

Feedback = Sign of the Jacobian

- A term in f(x) that depends on x is a feedback loop.
- Negative feedback builds stable points: homeostasis.
- Positive feedback builds switches and multistability.
- Fast negative + slow positive builds oscillation. Same fact as the Jacobian's sign.

Where You Now Stand

- Change has structure: fixed points, stability, bifurcations; and multistability, oscillation, chaos.
- Approximation is Taylor's theorem: smoothness makes the world locally linear.
- Your next real wall will usually be a nonlinear-dynamics wall; now you have its name.

Next: Chapter 11. A system can be determined yet unpredictable, and one fixed point can govern many mechanisms. Probability and universality take that step.